MULTIPLE CHOICE

1. Which of the following is a polynomial function?

A $y = -4x^4 + 4x^3 - 7x^2 + 9x$

B $f(x) = -4^x - 7$

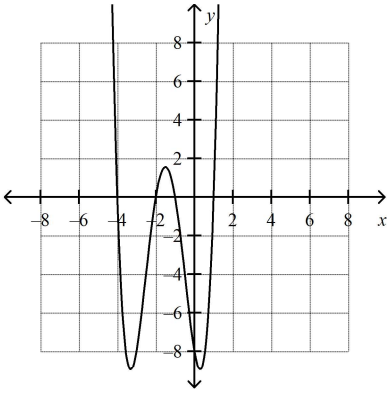
C $g(x) = \sqrt{x+4}$

D $y = \dfrac{-4x+9}{x^2}$

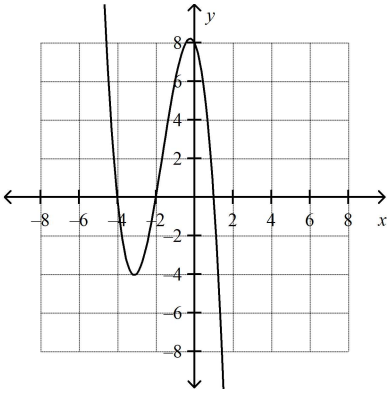
ANS: A DIF: Easy OBJ: Section 3.1 NAT: RF12
TOP: Characteristics of Polynomial Functions KEY: polynomial function

2. Which graph represents an odd-degree polynomial function with two x-intercepts?

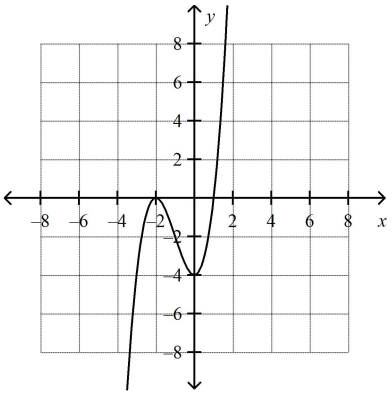
A



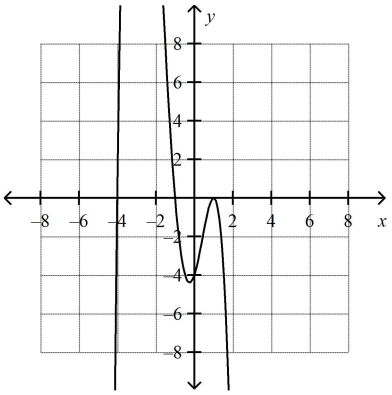
C



B



D

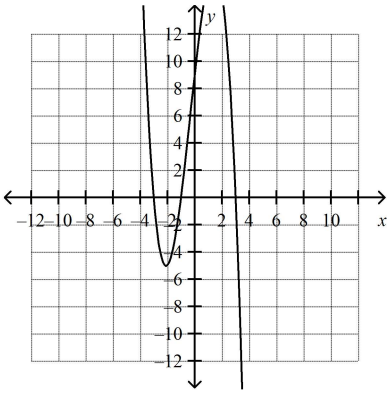


ANS: B DIF: Average OBJ: Section 3.1 NAT: RF12
TOP: Characteristics of Polynomial Functions KEY: odd-degree | x-intercepts

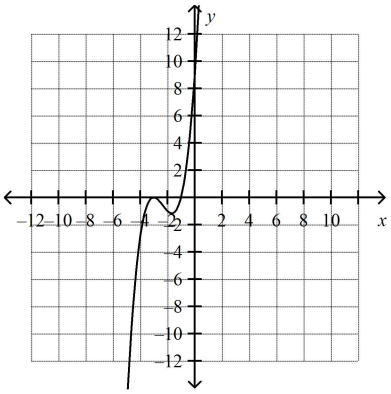
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3. Which graph represents an even-degree polynomial function with a y-intercept of 9?

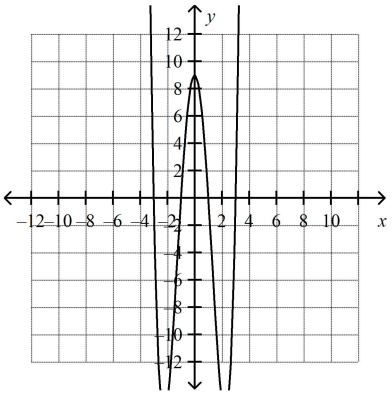
A



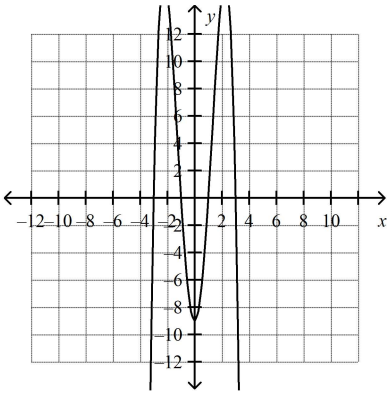
C



B



D



ANS: B DIF: Average OBJ: Section 3.1 NAT: RF12
TOP: Characteristics of Polynomial Functions
KEY: even-degree | x-intercepts | y-intercept

4. How many x-intercepts are possible for the polynomial function $h(x) = ax^5 + bx^4 + cx^3$?

A 4
B 5
C 3
D 1

ANS: B DIF: Easy OBJ: Section 3.1 NAT: RF12
TOP: Characteristics of Polynomial Functions KEY: x-intercepts

5. What is the restriction on x if $x^3 - 8x^2 + x + 5$ is divided by $x - 2$?

A $x \neq 8$
B $x \neq -5$
C $x \neq -1$
D $x \neq 2$

ANS: D DIF: Easy OBJ: Section 3.2 NAT: RF11
TOP: The Remainder Theorem KEY: restriction

6. If $-9x^3 + 9x^2 + 4$ is divided by $6x + 5$, then the restriction on x is

A $x \neq \frac{6}{5}$
B $x \neq \frac{5}{6}$
C $x \neq -\frac{5}{6}$
D $x \neq -\frac{6}{5}$

ANS: C DIF: Average OBJ: Section 3.2 NAT: RF12
TOP: The Remainder Theorem KEY: restriction

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7. What is the remainder when $x^4 - 8x^2 + 9x + 8$ is divided by $x + 6$?

A 1070
B –1070
C 962
D –962

ANS: C DIF: Average OBJ: Section 3.2 NAT: RF11
TOP: The Remainder Theorem KEY: remainder theorem | remainder

8. If $-2x^3 - 6x^2 + 5x - 7$ is divided by $x - 7$ to give a quotient of $-2x^2 - 20x - 135$ and a remainder of –952, then which of the following is true?

A $(x - 7)(-2x^2 - 20x - 135) = -952$
B $-2x^3 - 6x^2 + 5x - 7 = (x - 7)(-2x^2 - 20x - 135) - 952$
C $(x - 7)(-2x^2 - 20x - 135) = 952$
D $-2x^3 - 6x^2 + 5x - 7 = (x - 7)(-2x^2 - 20x - 135) + 952$

ANS: B DIF: Average OBJ: Section 3.2 NAT: RF11
TOP: The Remainder Theorem KEY: quotient | remainder

9. If $P(x) = -8x^3 - 8x^2 - 8x - 8$ is divided by $x - 3$, what is the remainder?

A –160
B –320
C 160
D 320

ANS: B DIF: Average OBJ: Section 3.2 NAT: RF11
TOP: The Remainder Theorem KEY: remainder

10. If $P(x) = -4x^3 + 8x^2 - 8x - 3$ is divided by $x - 4$, what is the remainder?

A $P(x - 4)$
B $P(4)$
C $P(x + 4)$
D $P(-4)$

ANS: B DIF: Average OBJ: Section 3.2 NAT: RF11
TOP: The Remainder Theorem KEY: remainder theorem | remainder

11. When $P(x) = 5x^3 - 2x + 2$ is divided by $5x - 2$, the remainder is

A $x^2 + x + \frac{12}{5}$
B $P(-2) = -34$
C $P(5/2) = 601/8$
D $P(2/5) = 38/25$

ANS: D DIF: Difficult + OBJ: Section 3.2 NAT: RF11
TOP: The Remainder Theorem KEY: remainder theorem | remainder

12. For a polynomial $P(x)$, if $P(6) = 0$, then which of the following must be a factor of $P(x)$?

A $x^2 - 6$
B $x + 6$
C $x^2 + 6$
D $x - 6$

ANS: D DIF: Easy OBJ: Section 3.3 NAT: RF11
TOP: The Factor Theorem KEY: factor theorem | factor

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13. A factor of $x^3 + 16x^2 + 79x + 120$ is

A $x + 3$ **C** $x - 3$

B $x - 5$ **D** $x - 8$

ANS: A DIF: Average OBJ: Section 3.3 NAT: RF11
TOP: The Factor Theorem KEY: factor theorem | integral zero theorem | factor

14. Which of the following binomials is a factor of $x^3 + 12x^2 + 29x + 18$?

A $x - 2$ **C** $x - 1$

B $x - 9$ **D** $x + 2$

ANS: D DIF: Average OBJ: Section 3.3 NAT: RF11
TOP: The Factor Theorem KEY: factor theorem | integral zero theorem | factor

15. Which set of values for x should be tested to determine the possible zeros of $y = x^3 + 6x^2 - 10x + 35$?

A ±5, ±7, and ±35 **C** 1, 5, 7, 12, and 35

B ±1, ±5, ±7, and ±35 **D** 1, 5, 7, and 35

ANS: B DIF: Average OBJ: Section 3.3 NAT: RF11
TOP: The Factor Theorem KEY: integral zero theorem

16. Determine the value of k so that $x + 2$ is a factor of $x^3 + 10x^2 + 23x + k$.

A $k = -1$ **C** $k = 14$

B $k = -14$ **D** $k = 1$

ANS: C DIF: Average OBJ: Section 3.3 NAT: RF11
TOP: The Factor Theorem KEY: factor theorem | factor

17. Which of the following is the fully factored form of $x^3 + 2x^2 - 23x - 60$?

A $(x + 3)(x - 4)(x + 5)$ **C** $(x - 3)(x + 4)(x + 5)$

B $(x + 3)(x + 4)(x - 5)$ **D** $(x - 3)(x - 4)(x - 5)$

ANS: B DIF: Easy OBJ: Section 3.3 NAT: RF11
TOP: The Factor Theorem KEY: factored form | factor theorem | factor

18. The fully factored form of $x^4 + 10x^3 + 7x^2 - 162x - 360$ is

A $(x + 5)(x + 3)(x + 6)(x - 4)$ **C** $(x + 5)(x - 3)(x - 6)(x - 4)$

B $(x - 5)(x + 3)(x + 6)(x - 6)$ **D** $x(x - 5)(x - 3)(x + 24)$

ANS: A DIF: Difficult OBJ: Section 3.3 NAT: RF11
TOP: The Factor Theorem KEY: factored form | factor theorem | factor

19. Which of the following is the fully factored form of $x^3 + 9x^2 - 4x - 36$?

A $(x - 2)^2(x + 9)$ **C** $(x + 2)(x - 2)(x - 9)$

B $(x - 2)^2(x - 9)$ **D** $(x + 2)(x - 2)(x + 9)$

ANS: D DIF: Average OBJ: Section 3.3 NAT: RF11
TOP: The Factor Theorem KEY: factored form | factor theorem | factor

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20. One root of the equation $x^3 + 7x^2 - 33x - 135 = 0$ is

A –3
B 3
C 9
D –5

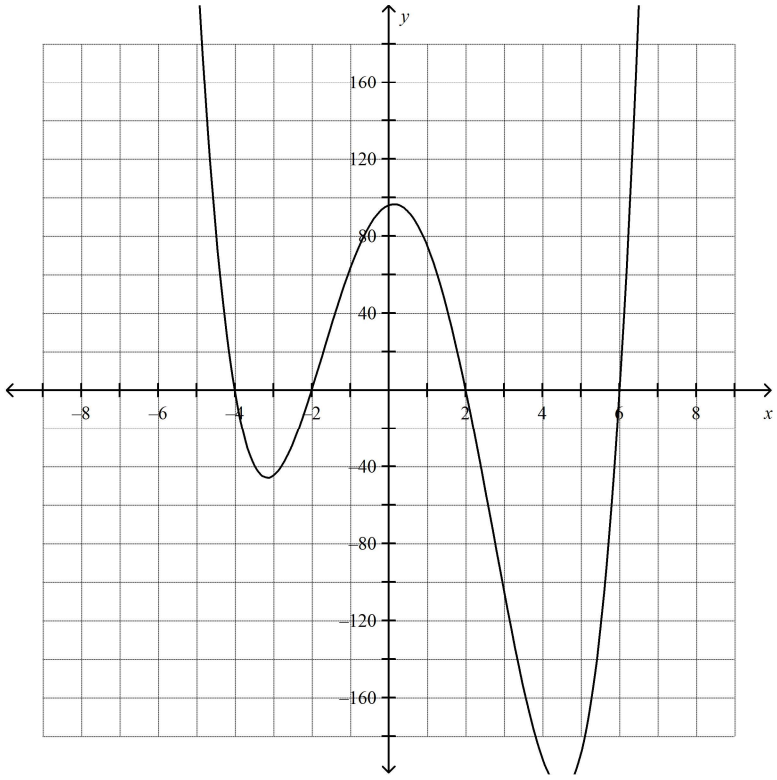
ANS: A DIF: Average OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions KEY: polynomial equation | roots

21. What is the maximum number of real distinct roots that a cubic equation can have?

A infinitely many
B 4
C 3
D 2

ANS: C DIF: Easy OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions KEY: polynomial equation | roots

22. Based on the graph of $f(x) = x^4 - 2x^3 - 24x^2 + 8x + 96$, what are the real roots of $x^4 - 2x^3 - 24x^2 + 8x + 96 = 0$?



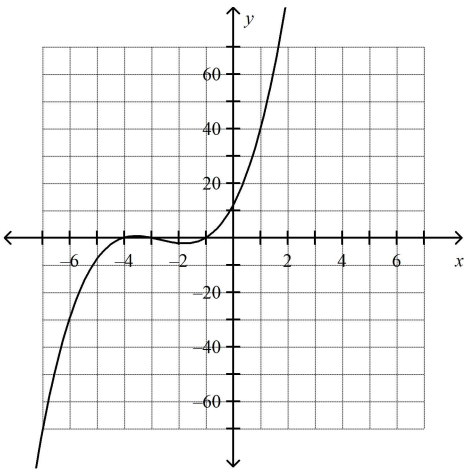
A –6, –2, 2, 4
B 6, 2, –2, –4
C there are no real roots
D impossible to determine

ANS: B DIF: Easy OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions KEY: polynomial equation | roots

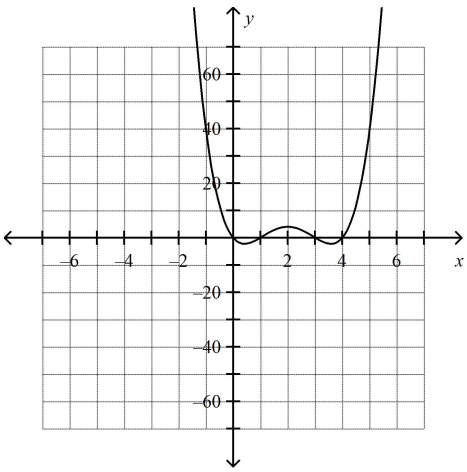
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23. Which of the following graphs of polynomial functions corresponds to a cubic polynomial equation with roots 4, 1, and 3?

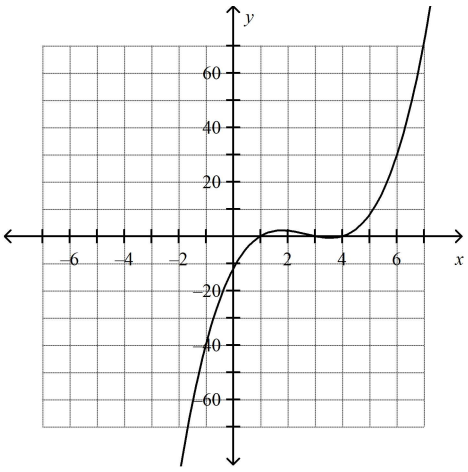
A



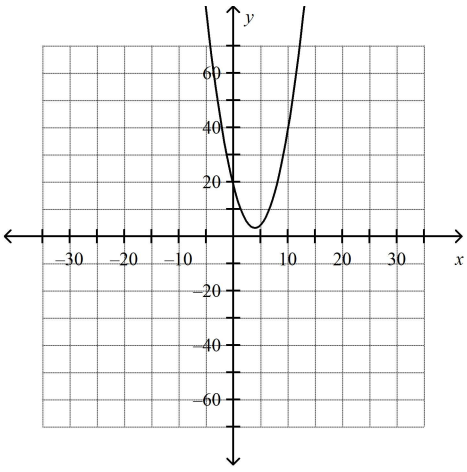
C



B



D

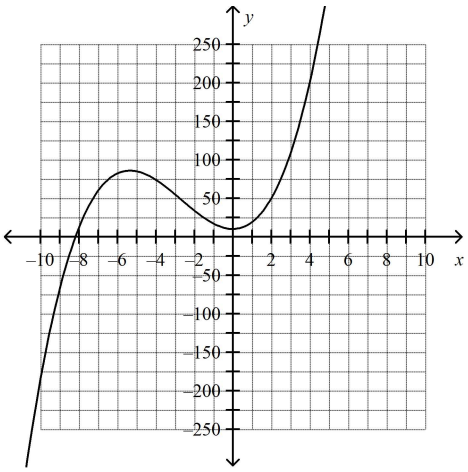


ANS: B DIF: Average OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions
KEY: polynomial equation | roots | graph

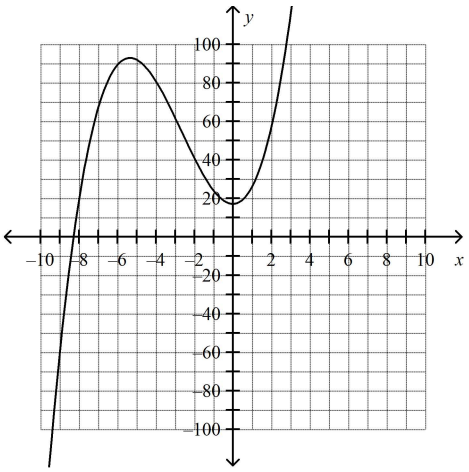
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24. The graph that corresponds to the function $f(x) = x^3 + 8x^2 + 17x + 10$ is

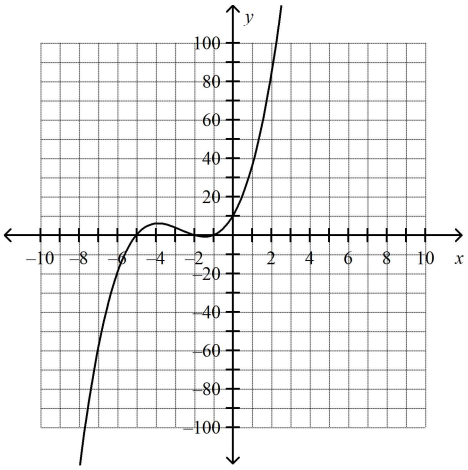
A



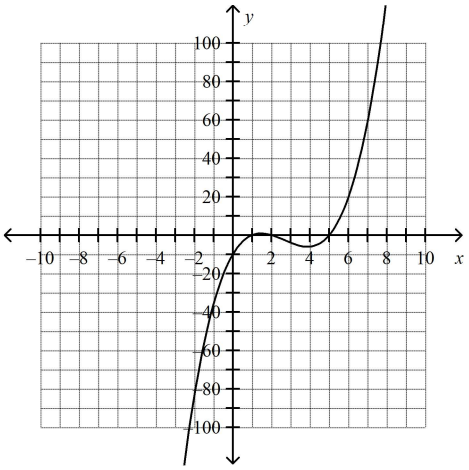
C



B



D

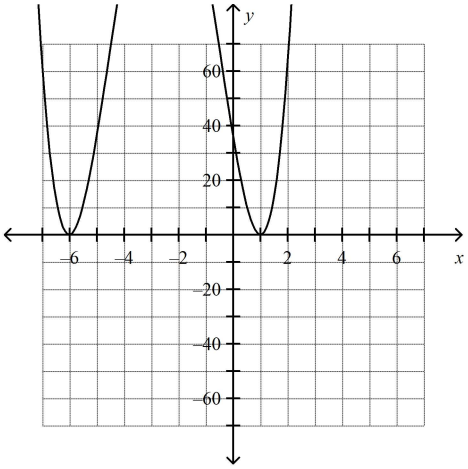


ANS: B DIF: Average OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions
KEY: polynomial equation | zeros | graph

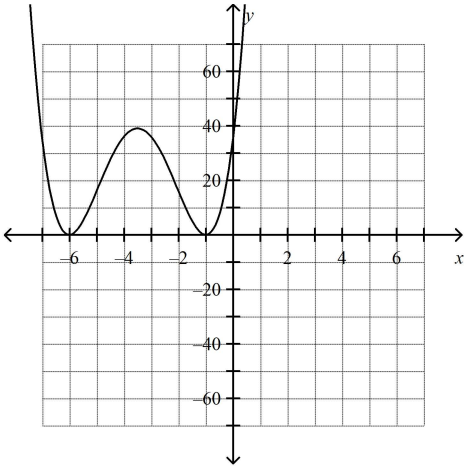
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25. Which of the following graphs of polynomial functions corresponds to a polynomial equation with zeros –6 (multiplicity of 2) and –1 (multiplicity of 2)?

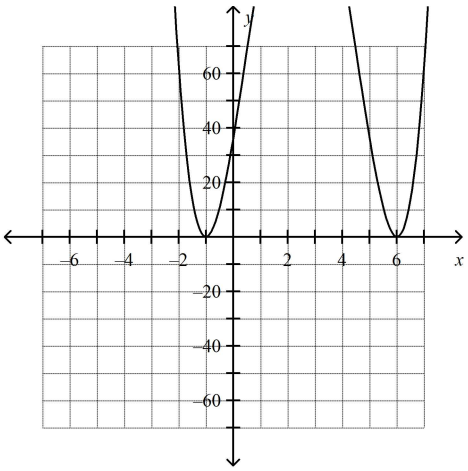
A



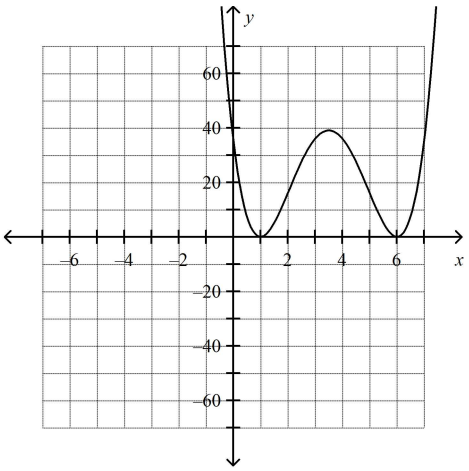
C



B



D



ANS: C DIF: Average OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions
KEY: polynomial equation | zeros | graph | multiplicity

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26. Given the function $y = x^3$, what are the parameters of the transformed function $y = \frac{1}{5}(x-2)^3 - 8$ and what is the effect of each parameter on the graph of the original function?

A $a = \frac{1}{5}$, vertical stretch about the *x*-axis by a factor of $\frac{1}{5}$
$h = -8$, horizontal translation 8 units right
$k = 2$, vertical translation 2 units down

B $a = 5$, vertical stretch about the *x*-axis by a factor of 5
$h = 2$, horizontal translation 2 units left
$k = -8$, vertical translation 8 units right

C $a = \frac{1}{5}$, vertical stretch about the *x*-axis by a factor of $\frac{1}{5}$
$h = 2$, horizontal translation 2 units right
$k = -8$, vertical translation 8 units down

D $a = 5$, vertical stretch about the *x*-axis by a factor of 5
$h = 2$, horizontal translation 2 units right
$k = -8$, vertical translation 8 units down

ANS: C DIF: Difficult OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions KEY: graph | transformations

COMPLETION

1. The function $f(x) = -x^3 - 5x^2 + 48x + 108$ is positive over the interval(s) _____.

ANS: $-2 < x < 6$ and $x < -9$

DIF: Difficult + OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions
KEY: characteristics of polynomials | inequality

2. The zeros of the function $f(x) = -x^3 + 18x^2 - 107x + 210$ are _____.

ANS: 5, 6, and 7

DIF: Difficult OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions
KEY: characteristics of polynomials | zeros

3. If $P(\frac{19}{18}) = 0$ for a polynomial $P(x)$, then _____ is a factor of $P(x)$.

ANS: $18x - 19$

DIF: Difficult + OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: factor theorem | rational zero theorem

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4. If $P(x) = -9x^3 - 3x^2 - 6x - 9$ is divided by $x - 4$, the remainder is _____.

ANS: –657

DIF: Average OBJ: Section 3.2 NAT: RF11 TOP: The Remainder Theorem
KEY: remainder theorem | remainder

5. The minimum number of *x*-intercepts that the graph of an odd-degree polynomial function can have is _____ .

ANS: 1

DIF: Easy OBJ: Section 3.1 NAT: RF11
TOP: Characteristics of Polynomial Functions KEY: x-intercepts | odd-degree

MATCHING

Match the correct term with the correct part of the statement.

$$\frac{x^4 - 16x^3 + 99x^2 - 234x + 216}{x + 8} = x^3 - 24x^2 + 291x - 2562 + \frac{20\,712}{x + 8}$$

A quotient
B remainder
C divisor
D dividend

1. $x^4 - 16x^3 + 99x^2 - 234x + 216$
2. $x + 8$
3. $x^3 - 24x^2 + 291x - 2562$
4. 20 712

1. ANS: D DIF: Easy OBJ: Section 3.2 NAT: RF11
TOP: The Remainder Theorem KEY: dividend
2. ANS: C DIF: Easy OBJ: Section 3.2 NAT: RF11
TOP: The Remainder Theorem KEY: divisor
3. ANS: A DIF: Easy OBJ: Section 3.2 NAT: RF11
TOP: The Remainder Theorem KEY: quotient
4. ANS: B DIF: Easy OBJ: Section 3.2 NAT: RF11
TOP: The Remainder Theorem KEY: remainder

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SHORT ANSWER

1. **a)** Use long division to divide $x^3 + 3x^2 - 7$ by $x + 2$. Express the result in quotient form.
b) Identify any restrictions on the variable.
c) Write the corresponding statement that can be used to check the division.
d) Verify your answer.

ANS:

a)
$$\begin{array}{r} x^2 + x - 2 \\ x+2\overline{)\,x^3 + 3x^2 + 0x - 7} \\ \underline{x^3 + 2x^2} \\ x^2 + 0x \\ \underline{x^2 + 2x} \\ -2x - 7 \\ \underline{-2x - 4} \\ -3 \end{array}$$

$$\frac{x^3 + 3x^2 - 7}{x+2} = x^2 + x - 2 - \frac{3}{x+2}$$

b) $x \neq -2$

c) $x^3 + 3x^2 - 7 = (x + 2)(x^2 + x - 2) - 3$

d) Expand the expression in part c) to verify.

$$\begin{aligned}(x+2)(x^2 + x - 2) - 3 &= x^3 + x^2 - 2x + 2x^2 + 2x - 4 - 3 \\ &= x^3 + 3x^2 - 7\end{aligned}$$

DIF: Average OBJ: Section 3.2 NAT: RF11 TOP: The Remainder Theorem
KEY: long division | restriction | quotient form | verify

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2. Show that $x + 1$ is a factor of $P(x) = x^3 + 2x^2 + 3x + 2$.

ANS:
Use the factor theorem. Find $P(-1)$.

$$P(x) = x^3 + 2x^2 + 3x + 2$$

$$P(-1) = (-1)^3 + 2(-1)^2 + 3(-1) + 2$$
$$= -1 + 2 - 3 + 2$$

$$P(-1) = 0$$

Since $P(-1) = 0$, $x + 1$ is a factor of $P(x)$.
Long division or synthetic division can also be used.

$$\begin{array}{r} x^2 + x + 2 \\ x+1 \overline{)\, x^3 + 2x^2 + 3x + 2} \\ \underline{x^3 + x^2} \\ x^2 + 3x \\ \underline{x^2 + x} \\ 2x + 2 \\ \underline{2x + 2} \\ 0 \end{array}$$

1	1	2	3	2
−		1	1	2
×	1	1	2	0

Since the remainder is 0, $x + 1$ is a factor of $P(x)$.

DIF: Easy OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: factor theorem | factor | remainder

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3. Factor fully.
a) $x^3 + 6x^2 + 11x + 6$
b) $4x^3 - 11x^2 - 3x$
c) $x^4 - 81$

ANS:
a) The possible factors are $(x \pm 1)$, $(x \pm 2)$, and $(x \pm 3)$. Try $x = -1$ using the factor theorem.

$$P(x) = x^3 + 6x^2 + 11x + 6$$

$$P(-1) = (-1)^3 + 6(-1)^2 + 11(-1) + 6$$
$$= -1 + 6 - 11 + 6$$

$$P(-1) = 0$$

Thus, $(x + 1)$ is a factor of $P(x)$.
Use synthetic division to find the quadratic factor.

1	1	6	11	6
−		1	5	6
×	1	5	6	0

Thus,

$$x^3 + 6x^2 + 11x + 6 = (x + 1)(x^2 + 5x + 6)$$
$$= (x + 1)(x + 2)(x + 3)$$

b) $4x^3 - 11x^2 - 3x = x(4x^2 - 11x - 3)$
$= x(4x^2 - 12x + x - 3)$
$= x[4x(x - 3) + (x - 3)]$
$= x(x - 3)(4x + 1)$

c) $x^4 - 81 = (x^2 - 9)(x^2 + 9)$
$= (x - 3)(x + 3)(x^2 + 9)$

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: factor theorem | factor NOT: A variety of factoring techniques is required.

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4. Factor fully.

a) $-n^3 - 4n^2 - 4n$

b) $12x^3 + 16x^2 - 5x - 3$

c) $2x^4 + x^3 - 10x - 5$

ANS:

a) $-n^3 - 4n^2 - 4n = -n(n^2 + 4n + 4)$

$= -n(n+2)^2$

b) Using the leading coefficient and the constant, the possible numbers to try in the factor theorem are $\pm\frac{1}{12}, \pm\frac{1}{6}, \pm\frac{1}{4}, \pm\frac{1}{3}, \pm\frac{1}{2}, \pm\frac{3}{4}, \pm 1$, and $\pm\frac{3}{2}$.

Try $x = \frac{1}{2}$.

$P(x) = 12x^3 + 16x^2 - 5x - 3$

$P\left(\frac{1}{2}\right) = 12\left(\frac{1}{2}\right)^3 + 16\left(\frac{1}{2}\right)^2 - 5\left(\frac{1}{2}\right) - 3$

$= \frac{12}{8} + \frac{16}{4} - \frac{5}{2} - 3$

$= \frac{3}{2} + 4 - \frac{5}{2} - 3$

$= -\frac{2}{2} + 1$

$P\left(\frac{1}{2}\right) = 0$

Thus, one factor is $x - \frac{1}{2}$.

Use synthetic division:

$-\frac{1}{2}$	12	16	–5	–3
–		–6	–11	–3
×	12	22	6	0

$12x^3 + 16x^2 - 5x - 3 = \left(x - \frac{1}{2}\right)(12x^2 + 22x + 6)$

$= (2x-1)(6x^2 + 11x + 3)$

$= (2x-1)(6x^2 + 2x + 9x + 3)$

$= (2x-1)[2x(3x+1) + 3(3x+1)]$

$= (2x-1)(3x+1)(2x+3)$

c) $2x^4 + x^3 - 10x - 5 = x^3(2x+1) - 5(2x+1)$

$= (2x+1)(x^3 - 5)$

DIF: Difficult + OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem

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KEY: factor theorem | factor | grouping

5. Factor fully.
a) $x^2(x-2)(x+2)+3x+6$
b) $16x^4-(x+1)^2$

ANS:
a) $x^2(x-2)(x+2)+3x+6=x^2(x-2)(x+2)+3(x+2)$

$$=(x+2)[x^2(x-2)+3]$$

$$=(x+2)(x^3-2x^2+3)$$

Use the factor theorem on the second factor. Try $x=-1$.

$P(x)=x^3-2x^2+3$

$P(-1)=(-1)^3-2(-1)^2+3$

$=-1-2+3$

$P(-1)=0$

Divide.

1	1	–2	0	3
–		1	–3	3
×	1	–3	3	0

The quotient is not factorable. Thus,

$x^2(x-2)(x+2)+3x+6=(x+2)(x+1)(x^2-3x+3)$

b) $16x^4-(x+1)^2=(4x^2)^2-(x+1)^2$

$$=[4x^2-(x+1)][4x^2+(x+1)]$$

$$=(4x^2-x-1)(4x^2+x+1)$$

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: factor theorem | factor | integral zero theorem | grouping | rational zero theorem
NOT: A variety of factoring techniques is required.

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6. Factor $2x^3 + 5x^2 - 14x - 8$ fully

ANS:

Possible values of x in the factor theorem are $\pm 1, \pm\frac{1}{2}, \pm 2, \pm 4$, and ± 8.

Try $x = 2$.

$$P(x) = 2x^3 + 5x^2 - 14x - 8$$

$$P(2) = 2(2)^3 + 5(2)^2 - 14(2) - 8$$

$$= 16 + 20 - 28 - 8$$

$$P(2) = 0$$

Thus, $x - 2$ is a factor of $P(x)$.

Divide.

-2	2	5	-14	-8
$-$		-4	-18	-8
$\times$	2	9	4	0

Thus,

$$2x^3 + 5x^2 - 14x - 8 = (x - 2)(2x^2 + 9x + 4)$$

$$= (x - 2)(2x^2 + x + 8x + 4)$$

$$= (x - 2)[x(2x + 1) + 4(2x + 1)]$$

$$= (x - 2)(2x + 1)(x + 4)$$

DIF: Difficult + OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: factor theorem | factor | integral zero theorem | grouping | rational zero theorem
NOT: A variety of factoring techniques is required.

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7. Solve by factoring.
a) $x^3 - 2x^2 - 5x + 6 = 0$
b) $x^3 - 3x^2 - 4x + 12 = 0$
c) $x^3 = 4x^2$

ANS:
a) The possible values of x for the factor theorem are ±1, ±2, ±3, and ±6. Try $x = 1$.

$$P(x) = x^3 - 2x^2 - 5x + 6$$

$$P(1) = 1^3 - 2(1)^2 - 5(1) + 6$$

$$= 1 - 2 - 5 + 6$$

$$P(1) = 0$$

Thus, $x - 1$ is a factor.
Find another factor using synthetic division.

–1	1	–2	–5	6
–		–1	1	6
×	1	–1	–6	0

Thus,

$$x^3 - 2x^2 - 5x + 6 = 0$$

$$(x - 1)(x^2 - x - 6) = 0$$

$$(x - 1)(x + 2)(x - 3) = 0$$

$$x = 1, -2, 3$$

b) $x^3 - 3x^2 - 4x + 12 = 0$

$$x^2(x - 3) - 4(x - 3) = 0$$

$$(x - 3)(x^2 - 4) = 0$$

$$(x - 3)(x - 2)(x + 2) = 0$$

$$x = 3, 2, -2$$

c) $x^3 = 4x^2$

$$x^3 - 4x^2 = 0$$

$$x^2(x - 4) = 0$$

$$x = 0, 4$$

DIF: Average OBJ: Section 3.3 | Section 3.4 NAT: RF11
TOP: The Factor Theorem | Equations and Graphs of Polynomial Functions
KEY: factor theorem | factor | integral zero theorem | roots

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8. Solve.

a) $3x^3 + 2x^2 - 8x + 3 = 0$

b) $2x^3 + x^2 - 10x - 5 = 0$

c) $5x^4 = 7x^2 - 2$

ANS:

a) Try $x = 1$ in the factor theorem.

$P(x) = 3x^3 + 2x^2 - 8x + 3$

$P(1) = 3(1)^3 + 2(1)^2 - 8(1) + 3$

$= 3 + 2 - 8 + 3$

$P(1) = 0$

Thus, $x - 1$ is a factor.

Use synthetic division to find another factor.

-1	3	2	-8	3
$-$		-3	-5	3
$\times$	3	5	-3	0

Another factor is $3x^2 + 5x - 3$. Thus,

$3x^3 + 2x^2 - 8x + 3 = 0$

$(x - 1)(3x^2 + 5x - 3) = 0$

$x = 1$ or $3x^2 + 5x - 3 = 0$

Use the quadratic formula to find the other solutions.

$3x^2 + 5x - 3 = 0$

$$x = \frac{-5 \pm \sqrt{5^2 - 4(3)(-3)}}{2(3)}$$

$$= \frac{-5 \pm \sqrt{61}}{6}$$

The solutions are $x = \frac{-5 - \sqrt{61}}{6}$, 1, $\frac{-5 + \sqrt{61}}{6}$.

b) $2x^3 + x^2 - 10x - 5 = 0$

$x^2(2x + 1) - 5(2x + 1) = 0$

$(x^2 - 5)(2x + 1) = 0$

$x = \sqrt{5}, -\sqrt{5}, -\frac{1}{2}$

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c) $5x^4 = 7x^2 - 2$

$$5x^4 - 7x^2 + 2 = 0$$

$$5x^4 - 5x^2 - 2x^2 + 2 = 0$$

$$5x^2(x^2 - 1) - 2(x^2 - 1) = 0$$

$$(x^2 - 1)(5x^2 - 2) = 0$$

$$x = -1, 1, -\sqrt{\frac{2}{5}}, \sqrt{\frac{2}{5}}$$

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: polynomial equation | factor theorem | factor | roots | integral zero theorem
NOT: A variety of factoring techniques is required.

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9. Solve by factoring.

a) $3x^4 - 48 = 0$

b) $-x^3 + 12x^2 - 47x + 60 = 0$

c) $21x^4 - 7x^3 - 6x^2 + 2x = 0$

ANS:

a)
$$3x^4 - 48 = 0$$
$$x^4 - 16 = 0$$
$$(x^2 - 4)(x^2 + 4) = 0$$
$$(x - 2)(x + 2)(x^2 + 4) = 0$$
$$x = -2, 2$$

b) Try $x = 3$ in the factor theorem.

$$P(x) = -x^3 + 12x^2 - 47x + 60$$
$$P(3) = -(3)^3 + 12(3)^2 - 47(3) + 60$$
$$= -27 + 108 - 141 + 60$$
$$P(3) = 0$$

Use synthetic division to find another factor.

-3	-1	12	-47	60
$-$		3	-27	60
$\times$	-1	9	-20	0

Thus,

$$-x^3 + 12x^2 - 47x + 60 = 0$$
$$(x - 3)(-x^2 + 9x - 20) = 0$$
$$-(x - 3)(x^2 - 9x + 20) = 0$$
$$-(x - 3)(x - 4)(x - 5) = 0$$
$$x = 3, 4, 5$$

c)
$$21x^4 - 7x^3 - 6x^2 + 2x = 0$$
$$x(21x^3 - 7x^2 - 6x + 2) = 0$$
$$x[7x^2(3x - 1) - 2(3x - 1)] = 0$$
$$x(3x - 1)(7x^2 - 2) = 0$$
$$x = 0, \frac{1}{3}, -\sqrt{\frac{2}{7}}, \sqrt{\frac{2}{7}}$$

DIF: Average OBJ: Section 3.3 | Section 3.4 NAT: RF11
TOP: The Factor Theorem | Equations and Graphs of Polynomial Functions
KEY: polynomial equation | factor theorem | factor | integral zero theorem | roots
NOT: A variety of factoring techniques is required.

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10. Solve by factoring.
a) $x^4 + 3x^2 - 28 = 0$
b) $2x^4 - 54x = 0$

ANS:

a) $x^4 + 3x^2 - 28 = 0$

$(x^2 + 7)(x^2 - 4) = 0$

$(x^2 + 7)(x - 2)(x + 2) = 0$

$x = 2, -2$

b) $2x^4 - 54x = 0$

$2x(x^3 - 27) = 0$

$2x(x - 3)(x^2 + 3x + 9) = 0$

$x = 0, 3$

DIF: Average OBJ: Section 3.3 | Section 3.4 NAT: RF11
TOP: The Factor Theorem | Equations and Graphs of Polynomial Functions
KEY: polynomial equation | factor theorem | factor | integral zero theorem | rational zero theorem | roots NOT: A variety of factoring techniques is required.

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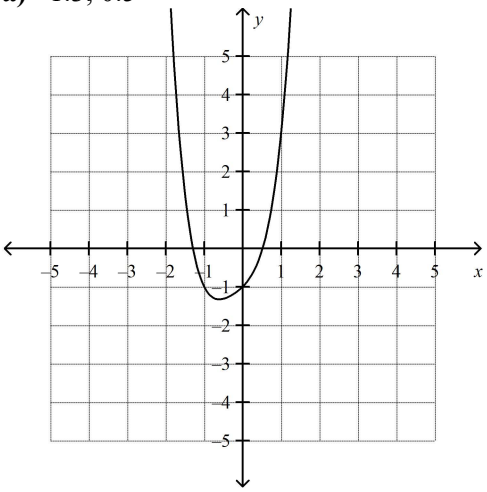
11. Solve by graphing using technology. Round the root(s) to one decimal place.

a) $x^4 + x^3 + x^2 + x = 1$

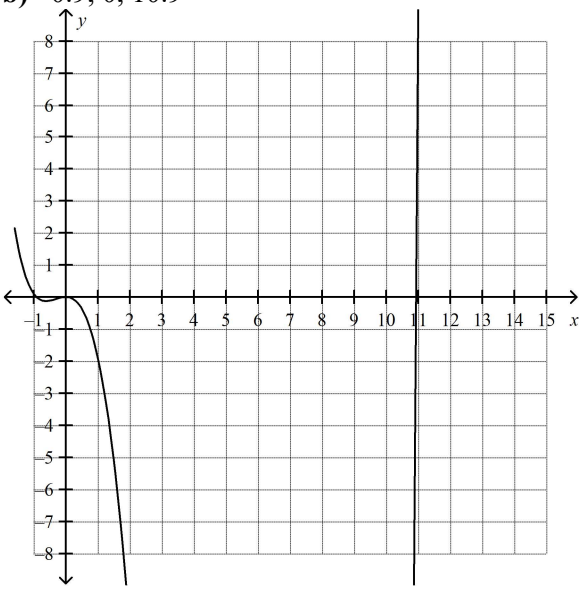
b) $0.1x^4 = x^3 + x^2$

ANS:

a) –1.3, 0.5



b) –0.9, 0, 10.9



DIF: Average OBJ: Section 3.4 NAT: RF11 | RF12
TOP: Equations and Graphs of Polynomial Functions
KEY: polynomial equation | technology | roots
NOT: Students will have to adjust the window for part b).

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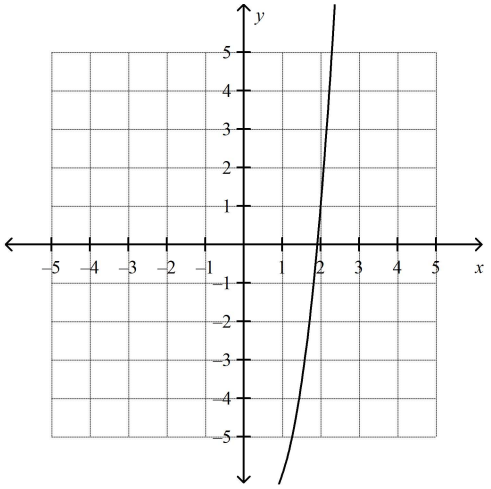
12. Solve by graphing using technology. Round answers to one decimal place.

a) $x^3 - 7 > 0$

b) $(x + 14)^3 \leq 1$

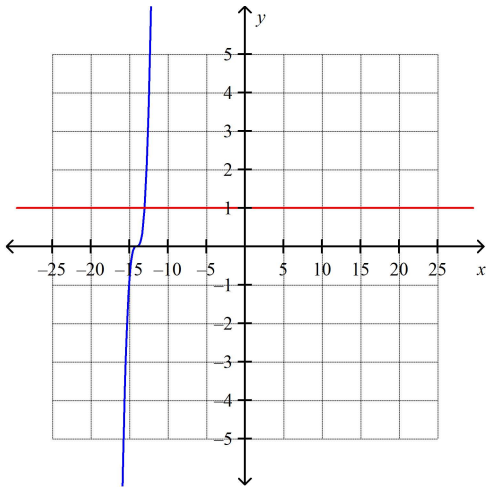
ANS:

a)



$x > 1.9$

b)



$x \leq -13$

DIF: Difficult + OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions
KEY: technology | inequality | graph | roots

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13. Solve the following inequalities using an algebraic method.

a) $x^2 + x - 2 > 0$

b) $x(x-2)(x-1)(x+1) \le 0$

c) $x^3 \ge 9x$

d) $-x^3 + 2x^2 + 4x - 8 \ge 0$

ANS:

a) $x^2 + x - 2 > 0$

$(x+2)(x-1) > 0$

When $x < -2$, both factors are less than zero, so their product is greater than zero. When $-2 < x < 1$, $x + 2 > 0$ and $x - 1 < 0$, so their product is less than zero. When $x > 1$, both factors are greater than zero, so their product is greater than zero. The solution to the inequality is $x < -2, x > 1$.

b) $x(x-2)(x-1)(x+1) \le 0$

Construct a table.

	$x < -1$	$-1 < x < 0$	$0 < x < 1$	$1 < x < 2$	$x > 2$
x	–	–	+	+	+
$x - 2$	–	–	–	–	+
$x - 1$	–	–	–	+	+
$x + 1$	–	+	+	+	+
$x(x-2)(x-1)(x+1)$	+	–	+	–	+

The expression is equal to zero at $x = -1$, $x = 0$, $x = 1$, and $x = 2$, so the solution is $-1 \le x \le 0, 1 \le x \le 2$.

c) $x^3 \ge 9x$

$x^3 - 9x \ge 0$

$x(x^2 - 9) \ge 0$

$x(x-3)(x+3) \ge 0$

Construct a table.

	$x < -3$	$-3 < x < 0$	$0 < x < 3$	$x > 3$
$x + 3$	–	+	+	+
x	–	–	+	+
$x - 3$	–	–	–	+
$x(x-3)(x+3)$	–	+	–	+

The expression is equal to zero at $x = -3$, $x = 0$, and $x = 3$, so the solution is $-3 \le x \le 0, x \ge 3$.

d) $-x^3 + 2x^2 + 4x - 8 \ge 0$

$-x^2(x-2) + 4(x-2) \ge 0$

$(x-2)(-x^2 + 4) \ge 0$

$(x-2)(x^2 - 4) \le 0$

$(x-2)(x-2)(x+2) \le 0$

Construct a table.

	$x < -2$	$-2 < x < 2$	$x > 2$
$x + 2$	–	+	+
$x - 2$	–	–	+
$x - 2$	–	–	+
$(x-2)(x-2)(x+2)$	–	+	+

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The expression is equal to zero at $x = 2$ and $x = -2$, so the solution is $x \leq -2, x = 2$.

DIF: Difficult + OBJ: Section 2.6 NAT: RF11
TOP: Polynomial and Rational Functions KEY: polynomial inequality

14. Determine the degree of the polynomial function (linear, quadratic, cubic, etc.) that the table of values represents.

x	*y*
–3	7
–2	9
–1	5
0	1
1	3
2	17
3	49

ANS:
Find the finite differences.

x	*y*	First Differences	Second Differences	Third Differences
–3	7			
–2	9	2		
–1	5	–4	–6	
0	1	–4	0	6
1	3	2	6	6
2	17	14	12	6
3	49	32	18	6

Since the third differences are constant, the polynomial is of degree three. It is a cubic.

DIF: Difficult + OBJ: Section 3.1 NAT: RF11
TOP: Characteristics of Polynomial Functions
KEY: table of values | polynomial function | finite differences | degree

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PROBLEM

1. The polynomial $6x^3 + mx^2 + nx - 5$ has a factor of $x + 1$. When divided by $x - 1$, the remainder is -4. What are the values of m and n?

ANS:
Let $P(x) = 6x^3 + mx^2 + nx - 5$.
By the factor theorem, since $x + 1$ is a factor of $P(x)$, then $P(-1) = 0$.

$$6(-1)^3 + m(-1)^2 + n(-1) - 5 = 0$$

$$-6 + m - n - 5 = 0$$

$$m - n = 11 \quad (1)$$

By the remainder theorem, when $P(x)$ is divided by $x - 1$, the remainder is $P(1)$.
Solve $P(1) = -4$.

$$6(1)^3 + m(1)^2 + n(1) - 5 = -4$$

$$6 + m + n - 5 = -4$$

$$m + n = -5 \quad (2)$$

Solve the system of equations. Add equation (1) to equation (2):

$$m - n = 11 \quad (1)$$

$$\underline{m + n = -5 \quad (2)}$$

$$2m = 6$$

$$m = 3$$

Substitute $m = 3$ into equation (2):

$$m + n = -5$$

$$3 + n = -5$$

$$n = -8$$

DIF: Difficult OBJ: Section 3.2 | Section 3.3 NAT: RF11
TOP: The Factor Theorem | The Remainder Theorem
KEY: factor theorem | remainder theorem | system of equations

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2. **a)** If $x + 1$ is a factor of $ax^4 + bx^2 + c$, what is the value of $a + b + c$?
b) Decide if $x + 1$ is a factor of each polynomial. Explain your reasoning.
i) $3x^4 + 3x^2 - 6$
ii) $17x^4 - 10x^2 - 7$
iii) $x^4 + x^2 + 1$

ANS:
a) Let $P(x) = ax^4 + bx^2 + c$.
By the factor theorem, since $x + 1$ is a factor of $P(x)$, then $P(-1) = 0$.

$$a(-1)^4 + b(-1)^2 + c = 0$$

$$a + b + c = 0$$

b) All the polynomials are of the form $ax^4 + bx^2 + c$.
So, from part a), $x + 1$ is a factor if $a + b + c = 0$.
i) For $3x^4 + 3x^2 - 6$, $a = 3$, $b = 3$, and $c = -6$. Since $a + b + c = 0$, $x + 1$ is a factor.
ii) For $17x^4 - 10x^2 - 7$, $a = 17$, $b = -10$, and $c = -7$. Since $a + b + c = 0$, $x + 1$ is a factor.
iii) For $x^4 + x^2 + 1$, $a = 1$, $b = 1$, and $c = 1$. Since $a + b + c \neq 0$, $x + 1$ is not a factor.

DIF: Average OBJ: Section 3.1 | Section 3.3 NAT: RF11
TOP: Characteristics of Polynomial Functions | The Factor Theorem
KEY: polynomial function | factor theorem

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3. Factor $2x^4 - 7x^3 - 41x^2 - 53x - 21$ fully.

ANS:
Let $P(x) = 2x^4 - 7x^3 - 41x^2 - 53x - 21$.
Test $x = -1$ in the factor theorem.

$$P(x) = 2x^4 - 7x^3 - 41x^2 - 53x - 21$$

$$P(-1) = 2(-1)^4 - 7(-1)^3 - 41(-1)^2 - 53(-1) - 21$$

$$= 2 + 7 - 41 + 53 - 21$$

$$P(-1) = 0$$

Thus, $x + 1$ is a factor. Divide to determine another factor.

1	2	–7	–41	–53	–21
–		2	–9	–32	–21
×	2	–9	–32	–21	0

Thus,

$$P(x) = (x + 1)(2x^3 - 9x^2 - 32x - 21)$$

Now factor the cubic. Test $x = -1$ in the factor theorem. Let $Q(x) = 2x^3 - 9x^2 - 32x - 21$.

$$Q(x) = 2x^3 - 9x^2 - 32x - 21$$

$$Q(-1) = 2(-1)^3 - 9(-1)^2 - 32(-1) - 21$$

$$= -2 - 9 + 32 - 21$$

$$Q(-1) = 0$$

Thus, $x + 1$ is a factor. Divide to determine another factor.

1	2	–9	–32	–21
–		2	–11	–21
×	2	–11	–21	0

Thus,

$$P(x) = (x + 1)(x + 1)(2x^2 - 11x - 21)$$

$$= (x + 1)^2(2x^2 - 14x + 3x - 21)$$

$$= (x + 1)^2[2x(x - 7) + 3(x - 7)]$$

$$= (x + 1)^2(x - 7)(2x + 3)$$

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: factor theorem | integral zero theorem | factor

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4. Factor $3x^4 + 5x^3 - 13x^2 - x + 6$ fully.

ANS:
Let $P(x) = 3x^4 + 5x^3 - 13x^2 - x + 6$.
Test $x = 1$ in the factor theorem.

$$P(x) = 3x^4 + 5x^3 - 13x^2 - x + 6$$

$$P(1) = 3(1)^4 + 5(1)^3 - 13(1)^2 - 1 + 6$$

$$= 3 + 5 - 13 - 1 + 6$$

$$P(1) = 0$$

Thus, $x - 1$ is a factor. Divide to determine another factor.

–1	3	5	–13	–1	6
–		–3	–8	5	6
×	3	8	–5	–6	0

Thus,

$$P(x) = (x - 1)(3x^3 + 8x^2 - 5x - 6)$$

Now factor the cubic. Test $x = 1$ in the factor theorem. Let $Q(x) = 3x^3 + 8x^2 - 5x - 6$.

$$Q(x) = 3x^3 + 8x^2 - 5x - 6$$

$$Q(1) = 3(1)^3 + 8(1)^2 - 5(1) - 6$$

$$= 3 + 8 - 5 - 6$$

$$Q(1) = 0$$

Thus, $x - 1$ is a factor. Divide to determine another factor.

–1	3	8	–5	–6
–		–3	–11	–6
×	3	11	6	0

Thus,

$$P(x) = (x - 1)(x - 1)(3x^2 + 11x + 6)$$

$$= (x - 1)^2(3x^2 + 9x + 2x + 6)$$

$$= (x - 1)^2[3x(x + 3) + 2(x + 3)]$$

$$= (x - 1)^2(x + 3)(3x + 2)$$

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Remainder Theorem
KEY: factor theorem | integral zero theorem | factor

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5. Show that $x + a$ is a factor of the polynomial $P(x) = (x + a)^4 + (x + c)^4 – (a – c)^4$.

ANS:
By the factor theorem, $x + a$ is a factor of $P(x)$ if $P(–a) = 0$.

$$P(–a) = (–a + a)^4 + (–a + c)^4 – (a – c)^4$$

$$= 0 + [–(a – c)]^4 – (a – c)^4$$

$$= (a – c)^4 – (a – c)^4$$

$$P(–a) = 0$$

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: factor theorem | factor

6. Prove that $x^2 + 5x + 6$ is a factor of $x^4 + 5x^3 + 2x^2 – 20x – 24$.

ANS:
The trinomial $x^2 + 5x + 6$ can be factored.
$x^2 + 5x + 6 = (x + 2)(x + 3)$
Check that these factors are also factors of $P(x) = x^4 + 5x^3 + 2x^2 – 20x – 24$ by using the factor theorem.

$$P(x) = x^4 + 5x^3 + 2x^2 – 20x – 24$$

$$P(–2) = (–2)^4 + 5(–2)^3 + 2(–2)^2 – 20(–2) – 24$$

$$= 16 – 40 + 8 + 40 – 24$$

$$P(–2) = 0$$

$$P(–3) = (–3)^4 + 5(–3)^3 + 2(–3)^2 – 20(–3) – 24$$

$$= 81 – 135 + 18 + 60 – 24$$

$$P(–3) = 0$$

Since $x + 2$ and $x + 3$ are both factors of $P(x)$, then their product, $x^2 + 5x + 6$, is also a factor of $P(x)$.

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: factor theorem | factor

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7. Given that –2 is a root of $x^3 + x = -4x^2 + 6$, find the other root(s).

ANS:
Rewrite in the form $P(x) = 0$.
$x^3 + 4x^2 + x - 6 = 0$
Since –2 is a root, $x + 2$ is a factor of $P(x)$. Divide to find the other factor.

2	1	4	1	–6
–		2	4	–6
×	1	2	–3	0

$$x^3 + 4x^2 + x - 6 = 0$$

$$(x+2)(x^2 + 2x - 3) = 0$$

$$(x+2)(x+3)(x-1) = 0$$

$$x = -2, -3, 1$$

Thus, the other roots are –3 and 1.

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Remainder Theorem
KEY: factor theorem | root

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8. Solve $2x^4 - x^3 = 7x^2 - 9x + 3$.

ANS:
Rewrite in the form $P(x) = 0$.
$2x^4 - x^3 - 7x^2 + 9x - 3 = 0$
Try $x = 1$ in the factor theorem.

$$P(x) = 2x^4 - x^3 - 7x^2 + 9x - 3$$

$$P(1) = 2(1)^4 - 1^3 - 7(1)^2 + 9(1) - 3$$

$$= 2 - 1 - 7 + 9 - 3$$

$$P(1) = 0$$

Since $x = 1$ is a zero of $P(x)$, $x - 1$ is a factor. Divide to determine another factor.

–1	2	–1	–7	9	–3
–		–2	–1	6	–3
×	2	1	–6	3	0

$$2x^4 - x^3 - 7x^2 + 9x - 3 = 0$$

$$(x - 1)(2x^3 + x^2 - 6x + 3) = 0$$

Now factor the cubic. Try $x = 1$ in the factor theorem. Let $Q(x) = 2x^3 + x^2 - 6x + 3$.

$$Q(x) = 2x^3 + x^2 - 6x + 3$$

$$Q(1) = 2(1)^3 + (1)^2 - 6(1) + 3$$

$$= 2 + 1 - 6 + 3$$

$$Q(1) = 0$$

Thus, $x - 1$ is a factor of $Q(x)$. Use division to find another factor.

–1	2	1	–6	3
–		–2	–3	3
×	2	3	–3	0

$$2x^4 - x^3 - 7x^2 + 9x - 3 = 0$$

$$(x - 1)(2x^3 + x^2 - 6x + 3) = 0$$

$$(x - 1)(x - 1)(2x^2 + 3x - 3) = 0$$

Thus, $x = 1$ or $2x^2 + 3x - 3 = 0$. Use the quadratic formula to find the other roots.

$$x = \frac{-3 \pm \sqrt{3^2 - 4(2)(-3)}}{2(2)}$$

$$= \frac{-3 \pm \sqrt{33}}{4}$$

The solution is $x = 1$, $x = \frac{-3 + \sqrt{33}}{4}$, or $x = \frac{-3 - \sqrt{33}}{4}$.

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Remainder Theorem
KEY: factor theorem | integral zero theorem | root

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9. The height of a square-based box is 4 cm more than the side length of its square base. If the volume of the box is 225 cm^3, what are its dimensions?

ANS:
Let x represent the side length of the base. Then, $V(x) = x^2(x + 4)$.
Solve $x^2(x + 4) = 225$.
Since 25 is a factor of 225, and 25 is a square, try $x = 5$ as a solution.

$$\text{L.S.} = x^2(x+4)$$
$$= 5^2(5+4)$$
$$= 25(9)$$
$$= 225$$
$$= \text{R.S.}$$

Thus, $x = 5$ is a solution. The dimensions of the box could be 5 cm by 5 cm by 9 cm.
Rewrite the equation in the form $P(x) = 0$.

$$x^2(x+4) = 225$$
$$x^3 + 4x^2 - 225 = 0$$

Since $x = 5$ is a solution, $x - 5$ is a factor of $P(x)$. Divide to find another factor.

–5	1	4	0	–225
–		–5	–45	–225
×	1	9	45	0

$$x^3 + 4x^2 - 225 = 0$$
$$(x-5)(x^2 + 9x + 45) = 0$$

$x^2 + 9x + 45 = 0$ has no real solutions. So the only solution is $x = 5$. The dimensions of the box are 5 cm by 5 cm by 9 cm.

DIF: Average OBJ: Section 3.3 | Section 3.4 NAT: RF11
TOP: The Factor Theorem | Equations and Graphs of Polynomial Functions
KEY: factor theorem | integral zero theorem | polynomial equation | root

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10. The height, h, in metres, of a weather balloon above the ground after t seconds can be modelled by the function $h(t) = -2t^3 + 3t^2 + 149t + 410$, for $0 \le t \le 10$. When is the balloon exactly 980 m above the ground?

ANS:
Solve $-2t^3 + 3t^2 + 149t + 410 = 980$.
Rewrite in the form $P(t) = 0$.
$2t^3 - 3t^2 - 149t + 570 = 0$
Try $x = 5$ in the factor theorem.
$P(t) = 2t^3 - 3t^2 - 149t + 570$

$P(5) = 2(5)^3 - 3(5)^2 - 149(5) + 570$

$= 250 - 75 - 745 + 570$

$P(5) = 0$
Divide to find another factor.

–5	2	–3	–149	570
–		–10	–35	570
×	2	7	–114	0

Thus,

$2t^3 - 3t^2 - 149t + 570 = 0$

$(t - 5)(2t^2 + 7t - 114) = 0$

$(t - 5)(2t^2 + 19t - 12t - 114) = 0$

$(t - 5)[t(2t + 19) - 6(2t + 19)] = 0$

$(t - 5)(2t + 19)(t - 6) = 0$

Since the time cannot be negative, the balloon is at exactly 980 m above the ground after 5 s and after 6 s.

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: roots | polynomial equation | factor | factor theorem | integral zero theorem | extraneous roots

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11. Amit has designed a rectangular storage unit to hold large factory equipment. His model has dimensions 1 m by 2 m by 4 m. By what amount should he increase each dimension to produce an actual storage unit that is 9 times the volume of his model?

ANS:
Let x represent the increase in each dimension. Then $V(x) = (1 + x)(2 + x)(4 + x)$.
Solve $(1 + x)(2 + x)(4 + x) = 9(1)(2)(4)$.

$$(1+x)(2+x)(4+x) = 9(1)(2)(4)$$

$$(2+3x+x^2)(4+x) = 72$$

$$8+2x+12x+3x^2+4x^2+x^3 = 72$$

$$x^3+7x^2+14x-64 = 0$$

Let $P(x) = x^3+7x^2+14x-64$, and try $x = 2$ in the factor theorem.

$$P(x) = x^3+7x^2+14x-64$$

$$P(2) = (2)^3+7(2)^2+14(2)-64$$

$$= 8+28+28-64$$

$$P(2) = 0$$

Thus, $x - 2$ is a factor of $P(x)$. Use division to find another factor.

–2	1	7	14	–64
–		–2	–18	–64
×	1	9	32	0

$$x^3+7x^2+14x-64 = 0$$

$$(x-2)(x^2+9x+32) = 0$$

$$x = 2 \text{ or } x^2+9x+32 = 0$$

$x^2+9x+32 = 0$ has no solution, so $x = 2$.
Each dimension should be increased by 2 m.

DIF: Average OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: factor theorem | integral zero theorem | factor | root

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12. Maria is designing a series of stationery trays for holding paper and envelopes. Each tray starts as a rectangular sheet of metal from which identical squares are cut from the four corners. The resulting rectangular flaps are then folded up and welded to produce an open-topped rectangular tray. What size square should Maria cut from a rectangular sheet of metal 15 cm by 25 cm in order to produce a tray with a volume of 476 cm^3? Round to the nearest hundredth of a centimetre, if necesary.

ANS:
Let x represent the side length of the square to be cut from each corner.
Then $V(x) = x(15 - 2x)(25 - 2x)$.
Solve $x(15 - 2x)(25 - 2x) = 476$.

$$x(15-2x)(25-2x) = 476$$

$$x(375 - 80x + 4x^2) = 476$$

$$4x^3 - 80x^2 + 375x - 476 = 0$$

Let $P(x) = 4x^3 - 80x^2 + 375x - 476$, and try $x = 4$ in the factor theorem.

$$P(x) = 4x^3 - 80x^2 + 375x - 476$$

$$P(4) = 4(4)^3 - 80(4^2) + 375(4) - 476$$

$$= 256 - 1280 + 1500 - 476$$

$$P(4) = 0$$

Thus, $x - 4$ is a factor of $P(x)$. Divide to find another factor.

–4	4	–80	375	–476
–		–16	256	–476
×	4	–64	119	0

Thus,

$$4x^3 - 80x^2 + 375x - 476 = 0$$

$$(x-4)(4x^2 - 64x + 119) = 0$$

$$x = 4 \text{ or } 4x^2 - 64x + 119 = 0$$

Use the quadratic formula to solve the quadratic equation.

$$x = \frac{64 \pm \sqrt{(-64)^2 - 4(4)(119)}}{2(4)}$$

$$= \frac{16 \pm \sqrt{137}}{2}$$

$$\approx 13.9 \text{ or } 2.15$$

$x = 13.9$ cm is not possible because of the starting dimensions of the sheet of metal. The side length of the square cut from each corner is either 4 cm or approximately 2.15 cm.

DIF: Difficult OBJ: Section 3.3 NAT: RF11 TOP: The Factor Theorem
KEY: factor theorem | integral zero theorem | factor | root | extraneous roots

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13. Solve $-x^3 + 5x^2 - 8x + 4 \geq 0$ algebraically and graphically.

ANS:
Multiply the equation by -1 to clear the negative leading coefficient.

$$-x^3 + 5x^2 - 8x + 4 \geq 0$$

$$-1(-x^3 + 5x^2 - 8x + 4) \leq 0$$

$$x^3 - 5x^2 + 8x - 4 \leq 0$$

Factor $x^3 - 5x^2 + 8x - 4$ using the factor theorem. Let $P(x) = x^3 - 5x^2 + 8x - 4$. Try $x = 1$ in the factor theorem.

$$P(x) = x^3 - 5x^2 + 8x - 4$$

$$P(1) = (1)^3 - 5(1)^2 + 8(1) - 4$$

$$= 1 - 5 + 8 - 4$$

$$P(1) = 0$$

Thus, $x - 1$ is a factor of $P(x)$. Divide to find another factor.

-1	1	-5	8	-4
$-$		-1	4	-4
$\times$	1	-4	4	0

Thus,

$$x^3 - 5x^2 + 8x - 4 \leq 0$$

$$(x - 1)(x^2 - 4x + 4) \leq 0$$

$$(x - 1)(x - 2)^2 \leq 0$$

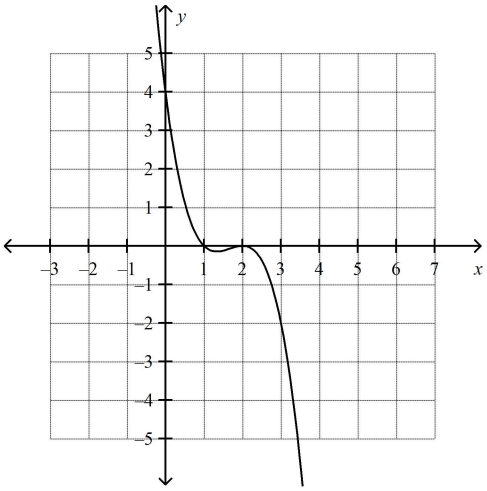
Construct a table.

	$x < 1$	$1 < x < 2$	$x > 2$
$x - 1$	$-$	$+$	$+$
$x - 2$	$-$	$-$	$+$
$x - 2$	$-$	$-$	$+$
$(x - 1)(x - 2)(x - 2)$	$-$	$+$	$+$

The expression is equal to zero at $x = 1$ and $x = 2$. Thus, the solution is $x \leq 1$ and $x = 2$.

Use a graphing calculator to graph the corresponding polynomial function, $y = -x^3 + 5x^2 - 8x + 4$. Then, use the **Zero** operation.

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The zeros are 1 and 2. From the graph, $-x^3 + 5x^2 - 8x + 4 \geq 0$ when $x \leq 1$ or $x = 2$.

DIF: Difficult + OBJ: Section 3.3 | Section 3.4 NAT: RF11 | RF12
TOP: The Factor Theorem | Equations and Graphs of Polynomial Functions
KEY: polynomial inequality | factor theorem | factor | integral zero theorem | root

14. Solve $3x^2(x^2 - 8) + 6x + 5 < 4x^4 - 6x(4x - 1) + 4$ algebraically. Display your result on a number line.

ANS:

$$3x^2(x^2 - 8) + 6x + 5 < 4x^4 - 6x(4x - 1) + 4$$

$$3x^4 - 24x^2 + 6x + 5 < 4x^4 - 24x^2 + 6x + 4$$

$$-x^4 + 1 < 0$$

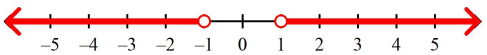
$$x^4 - 1 > 0$$

$$(x^2 + 1)(x^2 - 1) > 0$$

$$(x^2 + 1)(x - 1)(x + 1) > 0$$

$$(x - 1)(x + 1) > 0$$

Both sides of the equation can be divided by $x^2 + 1$ because it is always positive. When $x < -1$, both factors are negative, so their product is positive. When $-1 < x < 1$, $x - 1$ is negative and $x + 1$ is positive, so their product is negative. When $x > 1$, both factors are positive, so their product is positive.
Combining the results, the solution is $x < -1$ or $x > 1$.

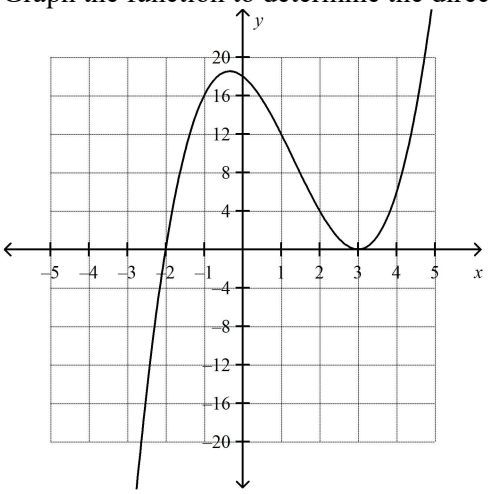


DIF: Difficult + OBJ: Section 3.3 | Section 3.4 NAT: RF11 | RF12
TOP: The Factor Theorem | Equations and Graphs of Polynomial Functions
KEY: factor theorem | number line | difference of squares | factor | root

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15. Create a cubic polynomial inequality for which $x = 3$ or $x \leq -2$ is the solution. Explain your reasoning.

ANS:
Answers may vary. Sample answer:
A cubic polynomial with a double root at $x = 3$ and another root at $x = -2$ would work, so try $f(x) = (x + 2)(x - 3)^2$.
Graph the function to determine the direction of the inequality sign.



When $x \leq -2$, $f(x) \leq 0$. When $x = 3$, $f(x) = 0$. So, the inequality is $f(x) \leq 0$, or $(x + 2)(x - 3)^2 \leq 0$.

DIF: Difficult + OBJ: Section 3.1 | Section 3.3 | Section 3.4
NAT: RF11
TOP: Characteristics of Polynomial Functions | The Factor Theorem | Equations and Graphs of Polynomial Functions KEY: polynomial function | graph | inequality | zeros

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16. Determine an equation for the polynomial function represented by the table of values.

x	*y*
–3	–30
–2	–10
–1	–2
0	0
1	2
2	10
3	30

ANS:
Calculate the first, second, and third differences.

x	*y*	First Differences	Second Differences	Third Differences
–3	–30			
–2	–10	20		
–1	–2	8	–12	
0	0	2	–6	6
1	2	2	0	6
2	10	8	6	6
3	30	20	12	6

Since the third differences are constant, the function is a cubic.
Let the function be of the form $f(x) = ax^3 + bx^2 + cx + d$.
Since $f(0) = 0$, $d = 0$. Thus, $f(x) = ax^3 + bx^2 + cx$.
Substitute other points in the table and solve for a, b, and c.

Substitute the point (–1, –2):

$f(x) = ax^3 + bx^2 + cx$

$-2 = a(-1)^3 + b(-1)^2 + c(-1)$

$-2 = -a + b - c \quad (1)$

Substitute the point (1, 2):

$f(x) = ax^3 + bx^2 + cx$

$2 = a(1)^3 + b(1)^2 + c(1)$

$2 = a + b + c \quad (2)$

Add equations (1) and (2):

$-2 = -a + b - c \quad (1)$

$2 = a + b + c \quad (2)$

$0 = 2b$

$b = 0$

Thus, the equation is of the form $f(x) = ax^3 + cx$. Substitute the points (2, 10) and (3, 30) to solve for a and c.

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$f(x) = ax^3 + cx$

$10 = 8a + 2c$

$5 = 4a + c \quad (3)$

$f(x) = ax^3 + cx$

$30 = 27a + 3c$

$10 = 9a + c \quad (4)$

Subtract equation (3) from equation (4):

$10 = 9a + c \quad (4)$

$5 = 4a + c \quad (3)$

$5 = 5a$

$a = 1$

Substitute $a = 1$ into equation (3):

$5 = 4a + c$

$5 = 4(1) + c$

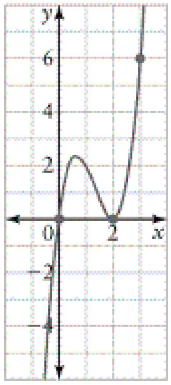
$c = 1$

Thus, the equation of the function is $f(x) = x^3 + x$.

DIF: Difficult + OBJ: Section 3.1 NAT: RF11
TOP: Characteristics of Polynomial Functions
KEY: table of values | polynomial function | system of equations | finite differences

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17. Determine an equation in expanded form for the polynomial function represented by the graph.



ANS:
The graph has a single zero at $x = 0$ and a double zero at $x = 2$. The graph also passes through the point (3, 6). Thus, the graph is of the form $y = ax(x - 2)^2$. Substitute the point (3, 6) to find a.

$$y = ax(x-2)^2$$

$$6 = a(3)(3-2)^2$$

$$6 = 3a$$

$$a = 2$$

Thus,

$$y = 2x(x-2)^2$$

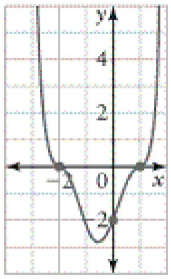
$$= 2x(x^2 - 4x + 4)$$

$$y = 2x^3 - 8x^2 + 8x$$

DIF: Average OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions
KEY: polynomial equation | graph | zeros

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18. Determine an equation in factored form for the polynomial function represented by the graph



ANS:
The graph has triple zeros at $x = -2$ and $x = 1$, and it passes through the point $(0, -2)$. Thus, the equation of the graph is of the form $y = a(x + 2)^3(x - 1)^3$. Substitute the point $(0, -2)$ to find a.

$$y = a(x+2)^3(x-1)^3$$

$$-2 = a(0+2)^3(0-1)^3$$

$$-2 = -8a$$

$$a = \frac{1}{4}$$

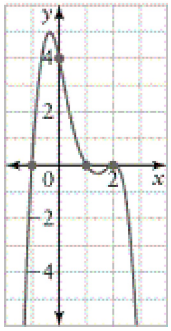
Thus,

$$y = \frac{1}{4}(x+2)^3(x-1)^3$$

DIF: Average OBJ: Section 3.4 NAT: RF11 | RF12
TOP: Equations and Graphs of Polynomial Functions
KEY: graph | polynomial equation | zeros

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19. Determine an equation in factored form for the polynomial function represented by the graph



ANS:
There are single zeros at $x = -1$ and $x = 1$, and a double zero at $x = 2$. The graph passes through the point (0, 4). Thus, the equation is of the form $y = a(x+1)(x-1)(x-2)^2$. Substitute the point (0, 4) to find a.

$y = a(x+1)(x-1)(x-2)^2$

$4 = a(0+1)(0-1)(0-2)^2$

$4 = -4a$

$a = -1$

Thus, $y = -(x+1)(x-1)(x-2)^2$.

DIF: Average OBJ: Section 3.4 NAT: RF12
TOP: Equations and Graphs of Polynomial Functions
KEY: graph | polynomial equation | zeros | y-intercept

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